

HADOOP COMMANDS

Explanation of HDFS Commands

1. `hdfs dfs -ls`

Purpose: Lists the files and directories stored in **HDFS (Hadoop Distributed File System)**.

Example

```
hdfs dfs -ls /
```

Explanation:

Displays the contents of the specified HDFS directory such as file name, size, permissions, and date.

2. `hdfs dfs -mkdir`

Purpose: Creates a new directory in HDFS.

Example

```
hdfs dfs -mkdir /input
```

Explanation:

Creates a directory named **input** in the HDFS root directory to store files.

3. `hdfs dfs -put`

Purpose: Uploads a file from the **local Linux system** to **HDFS**.

Example

```
hdfs dfs -put input.txt /input
```

Explanation:

Copies the file **input.txt** from the local system to the **/input** directory in HDFS.

4. `hdfs dfs -get`

Purpose: Downloads a file from **HDFS to the local system**.

Example

```
hdfs dfs -get /output/part-r-00000
```

Explanation:

Copies the file **part-r-00000** from HDFS to the current local directory.

5. `hdfs dfs -cat`

Purpose: Displays the contents of a file stored in HDFS.

Example

```
hdfs dfs -cat /input/input.txt
```

Explanation:

Shows the data stored inside the HDFS file directly in the terminal.

6. hdfs dfs -rm

Purpose: Deletes a file from HDFS.

Example

```
hdfs dfs -rm /input/input.txt
```

Explanation:

Removes the specified file from the Hadoop Distributed File System.

7. hdfs dfs -rm -r

Purpose: Deletes a directory and all its contents from HDFS.

Example

```
hdfs dfs -rm -r /input
```

Explanation:

Recursively removes the directory **input** and all files inside it.

Hadoop Service Commands (Cluster Control)

8. start-dfs.sh

Purpose: Starts the **HDFS** services.

Explanation:

It launches:

- NameNode
- DataNode
- SecondaryNameNode

These components manage and store distributed data in Hadoop.

9. start-yarn.sh

Purpose: Starts **YARN (Yet Another Resource Negotiator)** services.

Explanation:

It starts:

- ResourceManager
- NodeManager

These manage cluster resources and run MapReduce jobs.

10. jps

Purpose: Displays all running **Java processes**.

Explanation:

Used to verify whether Hadoop services are running.

Example output may show:

NameNode

DataNode

SecondaryNameNode

ResourceManager

NodeManager

Jps